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UNIVERSITI TEKNOLOGI MALAYSIA

**FACULTY OF COMPUTING**  
UTM Johor Bahru

## **SECJ 2203: Software Engineering**

Semester 01, 2024/2025

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### **PROJECT PROPOSAL**

### **COOPERATION SYSTEM**

**Lembaga Kemajuan Pertanian Kemubu (KADA)**  
**ORGANIZATION**

**Team Name: Group 5**

**Team Members:**

1. Muhammad Adam bin Razali
2. Muhammad Syahmi Faris bin Rusli
3. Muhammad Afiq Danish bin Mohd Hazni
4. Ahmad Ziyaad bin Mohd Abbas

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## 1. Introduction

The proposed Cooperation System aims to improve how Lembaga Kemajuan Pertanian Kemubu (KADA) Cooperation operates by solving inefficiencies in their current manual processes. Using the NABC framework, this introduction explains the Need (N), Approach (A), Benefit (B), and Competitor (C) of the system.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Need (N)     | At present, KADA Cooperation uses a manual, paper-based system to handle membership and loan applications. This approach is time-consuming, labor-intensive, and prone to errors, which causes frustration for both applicants and staff. The main issues include long application times, high operational costs, and the risk of losing physical documents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Approach (A) | <p>The new Cooperation System will introduce a digital platform that automates and simplifies KADA Cooperation's processes. The website for this system will be developed by using HTML, PHP, CSS (Bootstrap), and MySQL to ensure it is secure, user-friendly, and scalable. By following the System Development Life Cycle (SDLC) framework, the system will be designed with clear stages, including requirement gathering, system design, development, and testing.</p> <p>The platform will provide these functionalities such as:</p> <ol style="list-style-type: none"><li>1. User Authentication: Secure login for members and administrators.</li><li>2. Membership Applications: Easy online registration for cooperative membership.</li><li>3. Loan Application: Submission and tracking of loan requests digitally.</li><li>4. Data Management: A reliable database for securely storing and managing records.</li><li>5. Real-time tracking: Applicants can monitor their application status without visiting the office.</li></ol> |
| Benefit (B)  | <p>Some of the benefits switching to online system includes:</p> <ol style="list-style-type: none"><li>1. Improves User Experience: Customers can apply for accounts, submit documents, and receive approval from the comfort of their homes, eliminating the need to visit a branch location.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>2. Cost Savings: By automating the application process, KADA can reduce the need for in-person assistance and minimize the use of paper, leading to significant cost savings.</p> <p>3. Increased Efficiency: The digital system will streamline the application process, reducing the time and effort required from both customers and staff.</p>                                                                                                                                                                                                                                                                                                     |
| Competitor (C) | <p>There are other web-based systems available for cooperatives, but his system is specifically designed for KADA Cooperation. Unlike the others general cooperative platforms, it is tailored to fit their existing processes and includes features such as compliance with local regulations and support for specific financial services.</p> <p>This Project is an important step toward improving how KADA Cooperation operates. By addressing current issues with a well-planned web based system, it will enhance services, meet the needs of both members and staff, and help KADA Cooperation remain competitive in the cooperative industry.</p> |

## **2. Existing Systems**

### **a. Background of existing system**

The cooperation system used by KADA is currently being done manually. All of the forms and paperwork are submitted and stored in physical by the client and the organization. Applicants need to attend the KADA office in order to obtain the application form, and there is no computerized system for the submission of the documents required. The application type is divided into two, member application and loan application. If a client wants to apply for a loan, one must first become a member of KADA cooperation. The applicant needs to submit a member application and pay RM50 to become a member. When the application for membership is approved, then only they can proceed with loan application. Applicant's information is kept in a physical paper and the clerk must check their application one by one. Then, board members of KADA will conduct a meeting weekly to verify the application and decide whether to accept or reject the application. Once the result is finalized, the applicant will be notified through SMS messages. KADA offers a various type of loan for its member including Al-Bai, Al-Innah, Vehicle Restoration, Road Tax & Insurance and Al-Qadrul Hassan. Applicants can choose what type of loan they want to apply for according to their needs. The processes for loan application are quite similar to member application where applicant must come to the KADA office and obtain the application form, thus submit it with all the documents related at the office physically. A meeting between board members will be conducted also to determine the decision for the application. Lastly, successful applicants will be notified through SMS messages.

In addition, KADA's cooperation system includes a shares system which all members are eligible to join. Members can contribute to the organization's capital through purchasing of shares. These shares represent their ownership in the organization. Upon first time to apply for membership, the applicant needs to pay RM50. The registration fee will then be recorded as the applicant's shares when they are successful in becoming a member. Additionally, the shares amount will also put into factors which the member is eligible to apply for loan or not, all members must have at least RM350 worth of shares for them to apply for loan. If their shares do not fulfill the condition, they must first purchase the shares until they reach the required minimum amount.

## **b. Issues with existing system**

The issues arising from the manual handling of the application which are:

- i) **Data Security & Reliability** – All the related documents containing sensitive information about customers were stored physically. There might be risk of losing or leaking the data to malicious party because of human error. Moreover, the organization of the documents might not be in a systematic order, and it is possible for the data to mistakenly match with the wrong customer and end up providing misinformation about the customer. Thus, understanding the importance of having a convenient system in an organization might help the organization to provide secure and more reliable data for the customer.
- ii) **Timeliness** – The process of requiring and submitting documents is time consuming as the applicant needs to travel to KADA office multiple times. It requires a large amount of time just for the enquiries and submission of the documents. Some of the members might be living far away across the country and the distance between KADA office and their hometown must be put into consideration. To put it simply, the organization must provide a centralized system for the documentation which is accessible by the customer whenever or wherever they want.
- iii) **Scalability** – As time grows, KADA might experience an increasing number of its members. The manual system that is currently being used may struggle to handle a large volume of requests and applications in the future. Besides, the weekly meetings for applications approval might become a bottleneck and lead to slower response to the applicant, which will frustrate applicants. In summary, KADA needs to computerize their systems to give the best customer experience and enhance the efficiency of the system.

Table 1: New proposed system features

| Features                         | Description                                                                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Application for member           | Enables new users to submit an application for KADA membership                                                                                                                             |
| Application for loan             | Enables members to apply for loans through a digital interface, streamlining the submission, review, and approval process.                                                                 |
| Login                            | Offers safe user authentication so administrators and members can access their dashboards using their own login information.                                                               |
| Database Management System(DBMS) | Manages all backend data storage and retrieval operations, including member records, loan applications, and financial transactions, ensuring data integrity and accessibility using mysql. |
| Email notification               | Sends automated emails to members and administrators for critical updates, including application status, reminders, and approval confirmations.                                            |
| Shares system                    | Inform members about their shares amount and purchases                                                                                                                                     |
| Loan calculator                  | Enable users to calculate estimate total loan repayment by filling up the details required such as total desired loan, loan interest and loan duration                                     |

### 3. Proposed System

The proposed digital application portal for KADA Cooperation is a comprehensive system designed to modernize the organization's customer onboarding and application processes. This solution integrates seamlessly with KADA's broader ecosystem, including its financial management systems, customer database, and payment gateways. By ensuring compatibility with existing infrastructure, the system facilitates a smooth flow of information and functionality, reducing redundancies and enhancing operational efficiency. The centralized database is a key component, connecting various operational modules to automate tasks such as membership registration, loan application processing, and financial record management. Robust security protocols are incorporated to safeguard sensitive customer data, ensuring compliance with industry standards.

The system serves multiple user roles, each with specific functionalities tailored to their needs. Applicants, including prospective members or borrowers, can create accounts, apply for membership, and submit loan applications through an intuitive online portal. They are also equipped with tools to calculate loan repayment terms, track application statuses, receive real-time updates via email, and make online payments securely. KADA staff, such as clerks, are responsible for reviewing and verifying applications, updating user and loan information in the centralized database, and managing approvals based on eligibility criteria.

For more complex decision-making, KADA board members play a critical role in evaluating loan applications and approving or rejecting them based on risk assessments and policy guidelines. Additionally, system administrators ensure the smooth operation of the platform by managing technical issues, applying updates, maintaining security, and overseeing user access and permissions.

The portal boasts several key features, including a user-friendly interface designed for accessibility across all user roles, a secure centralized database for storing sensitive information, and a real-time loan calculator that simplifies financial planning for applicants. Automated email notifications enhance communication by updating users on their application status, payment confirmations, and other critical information. Furthermore, the integrated payment gateway supports various online payment methods, improving convenience for users.

The system's design emphasizes scalability and flexibility to accommodate KADA's growing user base and evolving needs. By implementing this digital application portal, KADA Cooperation seeks to enhance customer experience, streamline internal processes, and establish itself as a competitive player in the digital age. This initiative aligns with KADA's



strategic objectives of improving service delivery, reducing operational costs, and embracing innovative technologies.

#### 4. Software Process Model

The main goal of harnessing the software process model is to deliver a structured and clean approach for organizing, planning, and managing the development of a software system through a framework. It ensures that the project undergo a systematic development and any issues are tackled in a more professional way. To visualize, a software process model will assist in breaking down the project into a manageable stage which is call phases therefore making stakeholders, developers and users adapt to progress and deliverables of each phase with ease. Therefore, we are expecting to be able to accomplish:

**Organised Progression:** The project can be broken down into discrete stages using the model, each with its own goals and outputs. Stakeholders and team members can monitor progress and handle duties methodically with the aid of this segmentation.

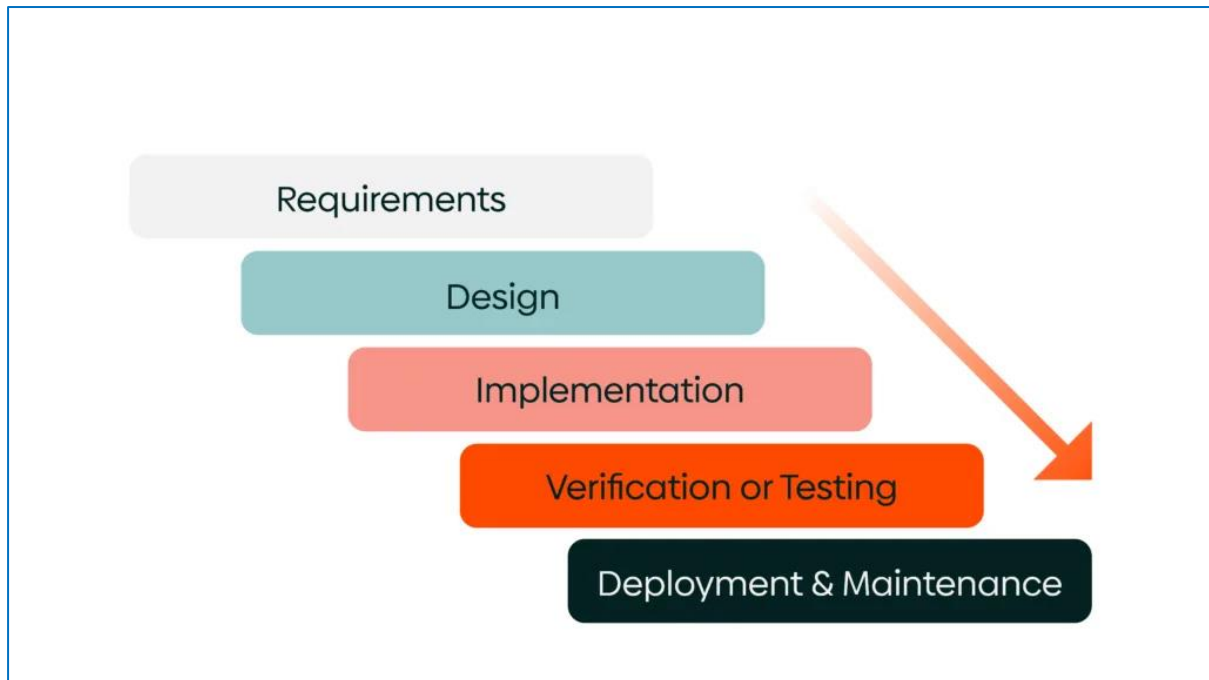
**Clear Communication:** It guarantees that developers, designers, testers, and clients, among other stakeholders, collaborate and communicate effectively. Before moving forward, each phase offers measurable results that may be examined and improved.

**Risk management:** By addressing any problems early on, such as during requirements collecting and design, the procedure lowers the possibility of expensive rework later on.

**Accountability and Transparency:** Better team accountability and stakeholder transparency are made possible by the clearly defined objectives and deliverables of each phase.

**Quality Assurance:** By using a methodical approach, the process model makes sure that the finished product satisfies

In this project, the Waterfall Model was selected due to its organized and sequential methodology that fits the project's needs for an orderly digital transformation of KADA's member application process. The traits of this model make it especially appropriate for projects where the requirements are well-defined and changes during development are anticipated to be minimal. This model specifically ensures that each phase starting from requirement gathering until testing are completed systematically before proceeding to the next phase. The waterfall model is depicted with the visual below:



*Figure 1: Waterfall Model Illustration*

In this project utilizing the Waterfall model, each stage generates important deliverables over a timeframe that extends beyond 14 weeks. In the Requirements Phase (weeks 1–3), the team collects and documents the necessary specifications, producing the Software Requirements Specification (SRS), use case diagrams, and a report validating the requirements. During the Design Phase (weeks 4–6), the system architecture, database schema, and UI prototypes are crafted, with deliverables that include the System Design Document (SDD) and Entity Relationship (ER) diagrams. The Implementation Phase (weeks 7–11) include coding and integrating the system, resulting in the complete source code, a functional software prototype, and a version control repository. Finally, the Testing Phase (weeks 12–14) confirms the functionality and quality of the software through comprehensive testing, with deliverables such as test cases, reports, and user acceptance approval. Although deployment and maintenance fall outside the course scope, they would encompass system

deployment, the creation of training materials, and a maintenance plan. This framework guarantees a methodical and high-quality development process.

## 5. Project Schedule

Illustrate the planned activities, duration and assigned person in-charge for the proposed project as a Gantt chart using any suitable project management tool.

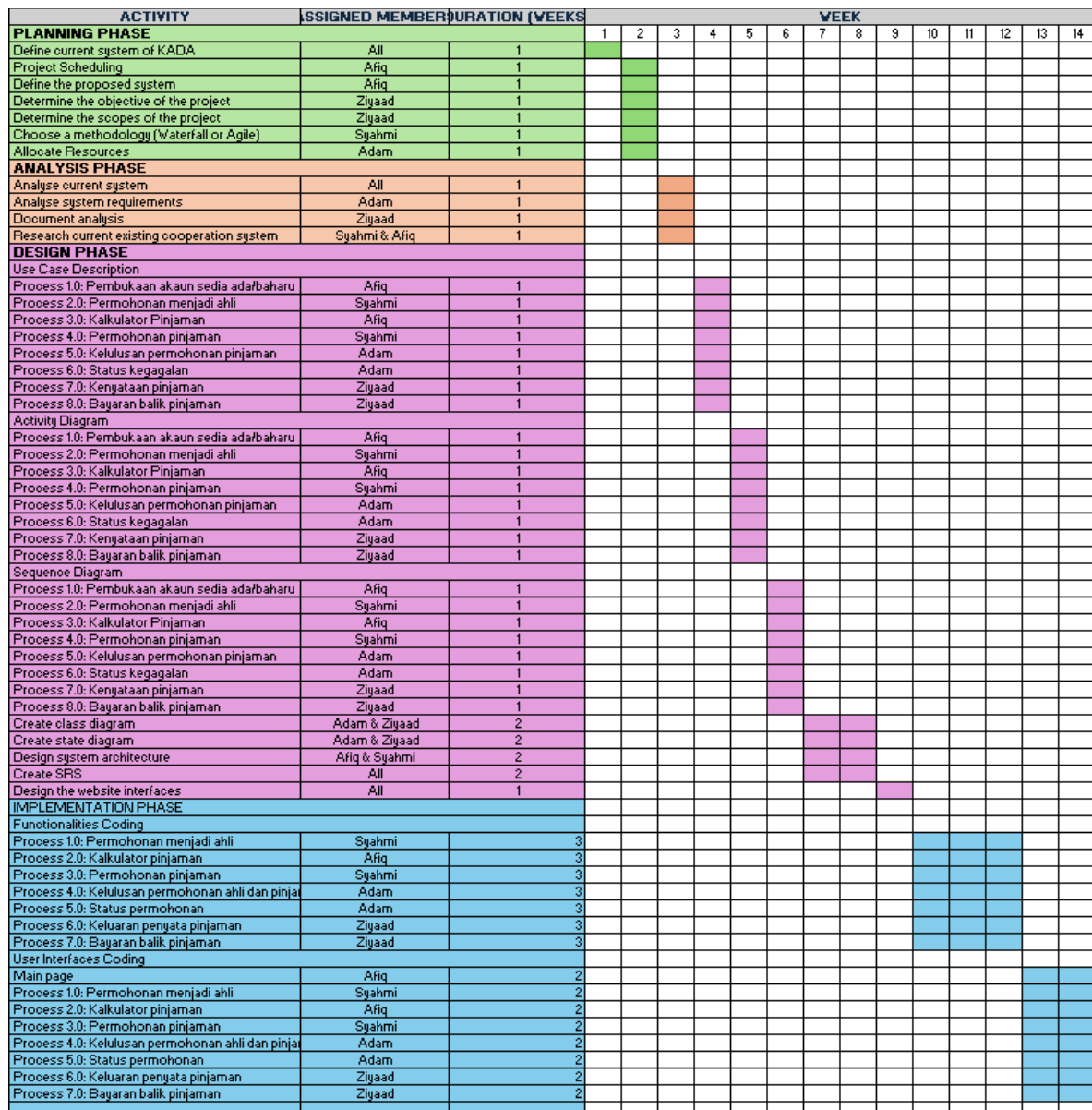


Figure 2: Project Gantt Chart

## **6. References**

[1] Sommerville, I. (2016). "Software Engineering", 10th Edition, US: Pearson.



